

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL 1

COURSE CODE: CSA16

COURSE NAME: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3, BL4, BL5)

Assessment Tool Weightage: 50%

QUESTIONS: 5

Total Marks:50

EXPERIMENTS:

1. Children of three ages are asked to indicate their preference for three photographs of adults. Do the data suggest that there is a significant relationship between age and photograph preference? What is wrong with this study? (10 Marks)

Age of child	Photograph:		
	A	B	C
5-6 years:	18	22	20
7-8 years:	2	28	40
9-10 years:	20	10	40

Use cov() to calculate the sample covariance between B and C.

Use another call to cov() to calculate the sample covariance matrix for the preferences.

Use cor() to calculate the sample correlation between B and C.

Use another call to cor() to calculate the sample correlation matrix for the preferences.

2. Assume the Tennis coach wants to determine if any of his team players are scoring outliers. To visualize the distribution of points scored by his players, then how can he decide to develop the box plot? (10 Marks)

Player ID	Player Name	Points Scored
P1	Player A	12
P2	Player B	15
P3	Player C	14
P4	Player D	10
P5	Player E	18
P6	Player F	20
P7	Player G	22
P8	Player H	13
P9	Player I	11
P10	Player J	35
P11	Player K	16
P12	Player L	17

P13	Player M	19
P14	Player N	21
P15	Player O	23

3. A bank wants to decide whether to approve a loan based on customer attributes like:

- Age
- Income
- Employment Status
- Credit Score

Using historical data, build a **Decision Tree classifier** to predict **Loan Approval (Yes/No)**.

(10 Marks)

Consider the Data Set

young,high,employed,good,yes
 young,high,self-employed,average,yes
 middle,high,employed,good,yes
 old,medium,employed,good,yes
 old,low,unemployed,poor,no
 old,low,self-employed,average,no
 middle,low,unemployed,poor,no
 young,medium,employed,average,yes
 young,low,unemployed,poor,no
 old,medium,self-employed,good,yes
 middle,medium,employed,average,yes
 middle,high,self-employed,good,yes
 old,medium,unemployed,poor,no
 young,medium,self-employed,average,yes
 middle,low,employed,poor,no

4. Two Maths teachers are comparing how their Year 9 classes performed in the end of year exams. Their results are as follows: (10 Marks)

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 84

Class B: 51, 56, 84, 60, 59, 70, 63, 66, 50

(i) Find which class had scored higher mean, median and range.

(ii) Plot above in boxplot and give the inferences

5. Suppose that a hospital tested the age and body fat data for 18 randomly selected adults with the following results: (10 Marks)

<i>age</i>	23	23	27	27	39	41	47	49	50
<i>%fat</i>	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2
<i>age</i>	52	54	54	56	57	58	58	60	61
<i>%fat</i>	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

(a) Calculate the mean, median, and standard deviation of *age* and *%fat*.

(b) Draw the boxplots for *age* and *%fat*.

(c) Draw a *scatter plot* and a *q-q plot* based on these two variables.

Answers:

1)

AIM: To calculate the **sample covariance and correlation** between photograph preferences and obtain their covariance and correlation matrices using R.

ALGORITHM:

1. Enter the preference data for photographs A, B and C.
2. Create vectors for A, B and C.
3. Create a data frame containing the preference values.
4. Use `cov(B,C)` to calculate covariance between B and C.
5. Use `cov()` on the complete data frame to calculate the covariance matrix.
6. Use `cor(B,C)` to calculate correlation between B and C.
7. Use `cor()` on the data frame to calculate the correlation matrix.
8. Display the results.

CODE:

```
A <- c(18, 2, 20)
B <- c(22, 28, 10)
C <- c(20, 40, 40)
preferences <- data.frame(A, B, C)
cat("Preference Data:\n")
print(preferences)
cat("\nSample Covariance between B and C:\n")
print(cov(B, C))
cat("\nSample Covariance Matrix:\n")
print(cov(preferences))
cat("\nSample Correlation between B and C:\n")
print(cor(B, C))
cat("\nSample Correlation Matrix:\n")
print(cor(preferences))
```

OUTPUT:

```
> A <- c(18, 2, 20)
> B <- c(22, 28, 10)
> C <- c(20, 40, 40)
> preferences <- data.frame(A, B, C)
> cat("Preference Data:\n")
Preference Data:
> print(preferences)
  A  B  C
1 18 22 20
2  2 28 40
3 20 10 40
> cat("\nSample Covariance between B and C:\n")

Sample Covariance between B and C:
> print(cov(B, C))
[1] -20
> cat("\nSample Covariance Matrix:\n")

Sample Covariance Matrix:
> print(cov(preferences))
      A      B      C
A  97.33333 -74 -46.66667
B -74.00000  84 -20.00000
C -46.66667 -20 133.33333
> cat("\nSample Correlation between B and C:\n")

Sample Correlation between B and C:
> print(cor(B, C))
[1] -0.1889822
> cat("\nSample Correlation Matrix:\n")

Sample Correlation Matrix:
> print(cor(preferences))
      A      B      C
A  1.0000000 -0.8183918 -0.4096440
B -0.8183918  1.0000000 -0.1889822
C -0.4096440 -0.1889822  1.0000000
```

RESULT:

The covariance between **B and C** is **-20** and their correlation is approximately **-0.189**. Thus, B and C have a **weak negative correlation**.

2)

AIM:

To visualize the distribution of points scored by tennis players using a **boxplot** and identify possible outliers.

ALGORITHM:

1. Enter the player names and their points scored.
2. Store the points in a vector.
3. Calculate the five-number summary using summary().
4. Create a boxplot using boxplot().
5. Identify outliers using boxplot.stats()\$out.
6. Display the outlier values.
7. Interpret the boxplot.

CODE:

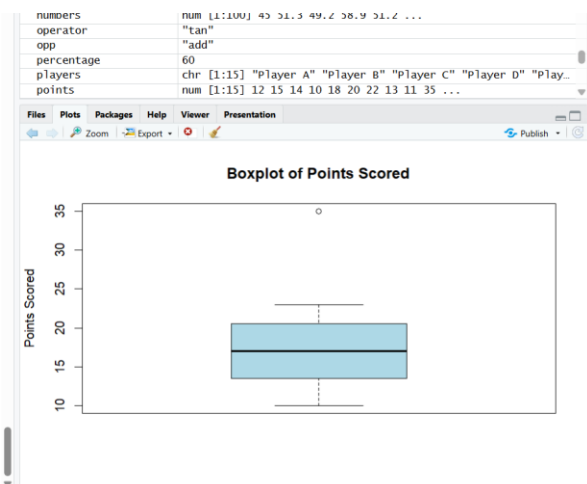
```
players <- c("Player A", "Player B", "Player C", "Player D", "Player E", "Player F", "Player G", "Player H", "Player I", "Player J", "Player K", "Player L", "Player M", "Player N", "Player O")
points <- c(12, 15, 14, 10, 18, 20, 22, 13, 11, 35, 16, 17, 19, 21, 23)
data <- data.frame(players, points)
cat("Player Scores:\n")
print(data)
boxplot(points, main = "Boxplot of Points Scored", ylab = "Points Scored", col = "lightblue")
cat("\nFive Number Summary:\n")
print(summary(points))
cat("\nOutliers:\n")
print(boxplot.stats(points)$out)
```

OUTPUT:

```
> players <- c("Player A", "Player B", "Player C", "Player D", "Player E", "Player F", "Player G", "Player H", "Player I", "Player J", "Player K", "Player L", "Player M", "Player N", "Player O")
> points <- c(12, 15, 14, 10, 18, 20, 22, 13, 11, 35, 16, 17, 19, 21, 23)
> data <- data.frame(players, points)
> cat("Player Scores:\n")
Player Scores:
> print(data)
  players points
1 Player A    12
2 Player B    15
3 Player C    14
4 Player D    10
5 Player E    18
6 Player F    20
7 Player G    22
8 Player H    13
9 Player I    11
10 Player J    35
11 Player K    16
12 Player L    17
13 Player M    19
14 Player N    21
15 Player O    23
> boxplot(points, main = "Boxplot of Points Scored", ylab = "Points Scored", col = "lightblue")
> cat("\nFive Number Summary:\n")

Five Number Summary:
> print(summary(points))
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  10.00  13.50   17.00  17.73  20.50   35.00
> cat("\nOutliers:\n")

Outliers:
> print(boxplot.stats(points)$out)
[1] 35
> |
```



RESULT:

The boxplot successfully identifies **35 points (Player J)** as an outlier.

3)

AIM:

To build a **Decision Tree classifier in R** for predicting whether a bank loan should be approved based on customer attributes.

ALGORITHM:

1. Create the given loan dataset.
2. Define Age, Income, Employment, Credit Score and Loan Approval.
3. Convert categorical variables into factors.
4. Load the rpart package.
5. Build the Decision Tree using rpart().
6. Visualize the Decision Tree using rpart.plot().
7. Predict loan approval for the given records.
8. Create a confusion matrix.
9. Calculate classification accuracy.
10. Display the result.

CODE:

```
library(rpart)
library(rpart.plot)
data <- data.frame(
  Age =
c("young","young","middle","old","old","old","middle","young","young","old","middle","mi-
ddle","old","young","middle"),
  Income =
c("high","high","high","medium","low","low","low","medium","low","medium","medium","
high","medium","medium","low"),
  Employment = c("employed","self-employed","employed","employed","unemployed","self-
employed","unemployed","employed","unemployed","self-employed","employed","self-
employed","unemployed","self-employed","employed"),
  Credit =
c("good","average","good","good","poor","average","poor","average","poor","good","average
","good","poor","average","poor"),
  Loan =
c("yes","yes","yes","yes","no","no","no","yes","no","yes","yes","yes","no","yes","no")
)
data$Age <- factor(data$Age)
data$Income <- factor(data$Income)
data$Employment <- factor(data$Employment)
data$Credit <- factor(data$Credit)
data$Loan <- factor(data$Loan)
model <- rpart(Loan ~ Age + Income + Employment + Credit,data = data,method =
"class",control = rpart.control(minsplit = 2, cp = 0))
print(model)
rpart.plot(model,main = "Decision Tree for Loan Approval")
predicted <- predict(model, data, type = "class")
cat("\nPredicted Loan Approval:\n")
print(predicted)
cat("\nConfusion Matrix:\n")
print(table(Actual = data$Loan,Predicted = predicted))
accuracy <- mean(predicted == data$Loan)
cat("\nAccuracy:\n")
print(accuracy)
cat("\nAccuracy Percentage:\n")
print(accuracy * 100)
```

OUTPUT:

```
> data$loan <- factor(data$loan)
> model <- rpart(loan ~ Age + Income + Employment + Credit, data = data, method = "class", control
= rpart.control(minsplit = 2, cp = 0))
> print(model)
n= 15

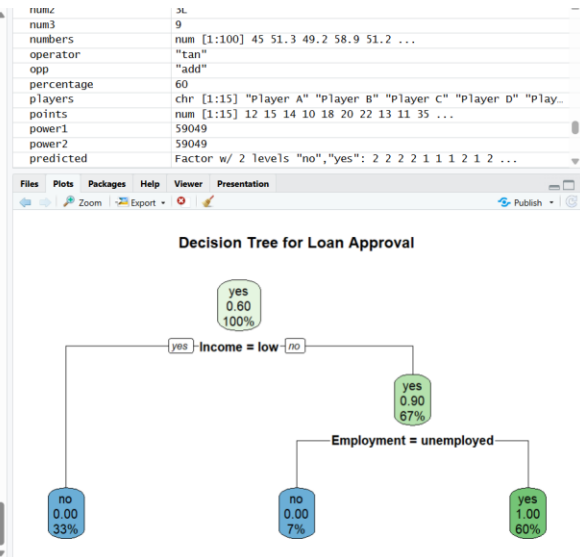
node), split, n, loss, yval, (yprob)
# denotes terminal node
1) root 15 6 yes (0.4000000 0.6000000)
  2) Income=low 5 0 no (1.0000000 0.0000000) *
    3) Income=high,medium 10 1 yes (0.1000000 0.9000000)
      6) Employment=unemployed 1 0 no (1.0000000 0.0000000) *
        7) Employment=employed,self-employed 9 0 yes (0.0000000 1.0000000) *
> rpart.plot(model, main = "Decision Tree for Loan Approval")
> predicted <- predict(model, data, type = "class")
> cat("\nPredicted Loan Approval:\n")

Predicted Loan Approval:
> print(predicted)
 1  2  3  4  5  6  7  8  9 10 11 12 13 14 15
yes yes yes yes no no yes no yes yes yes no yes no
Levels: no yes
> cat("\nConfusion Matrix:\n")

Confusion Matrix:
> print(table(Actual = data$loan, Predicted = predicted))
      Predicted
Actual no yes
      no  6  0
      yes 0  9
> accuracy <- mean(predicted == data$loan)
> cat("\nAccuracy:\n")

Accuracy:
> print(accuracy)
[1] 1
> cat("\nAccuracy Percentage:\n")

Accuracy Percentage:
> print(accuracy * 100)
[1] 100
>
```



RESULT:

The Decision Tree classifier successfully predicts the loan approval for the given training data with an accuracy of **100%**.

4)

AIM:

To calculate and compare the **mean, median and range** of two classes and visualize their score distributions using boxplots.

ALGORITHM:

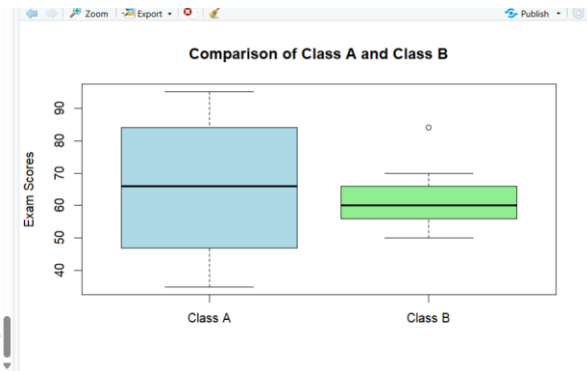
1. Enter the marks of Class A.
2. Enter the marks of Class B.
3. Calculate the mean of both classes.
4. Calculate the median of both classes.
5. Calculate the range of both classes.
6. Compare the results.
7. Plot the marks using a boxplot.
8. Interpret the boxplot.

CODE:

```
class_A <- c(76, 35, 47, 64, 95, 66, 89, 36, 84)
class_B <- c(51, 56, 84, 60, 59, 70, 63, 66, 50)
mean_A <- mean(class_A)
mean_B <- mean(class_B)
median_A <- median(class_A)
median_B <- median(class_B)
range_A <- max(class_A) - min(class_A)
range_B <- max(class_B) - min(class_B)
cat("Class A:\n")
cat("Mean =", mean_A, "\n")
cat("Median =", median_A, "\n")
cat("Range =", range_A, "\n")
cat("\nClass B:\n")
cat("Mean =", mean_B, "\n")
cat("Median =", median_B, "\n")
cat("Range =", range_B, "\n")
boxplot(class_A, class_B, names = c("Class A", "Class B"), main = "Comparison of Class A
and Class B", ylab = "Exam Scores", col = c("lightblue", "lightgreen"))
```

OUTPUT:

```
> class_A <- c(76, 35, 47, 64, 95, 66, 89, 36, 84)
> class_B <- c(51, 56, 84, 60, 39, 70, 63, 66, 50)
> mean_A <- mean(class_A)
> mean_B <- mean(class_B)
> median_A <- median(class_A)
> median_B <- median(class_B)
> range_A <- max(class_A) - min(class_A)
> range_B <- max(class_B) - min(class_B)
> cat("Class A:\n")
Class A:
> cat("Mean =", mean_A, "\n")
Mean = 65.77778
> cat("Median =", median_A, "\n")
Median = 66
> cat("Range =", range_A, "\n")
Range = 60
> cat("\nClass B:\n")
Class B:
> cat("Mean =", mean_B, "\n")
Mean = 62.11111
> cat("Median =", median_B, "\n")
Median = 60
> cat("Range =", range_B, "\n")
Range = 34
> boxplot(class_A, class_B, names = c("Class A", "Class B"), main = "Comparison of Class A and Class B", ylab = "Exam Scores", col = c("lightblue", "lightgreen"))
>
```



RESULT:

Class A scored higher in both mean and median. Class B has a smaller range, indicating that its scores are **more consistent**, whereas Class A has greater variation.

5)

AIM:

To calculate the **mean, median and standard deviation** of age and body fat and visualize the data using boxplots, scatter plot and Q-Q plots.

ALGORITHM:

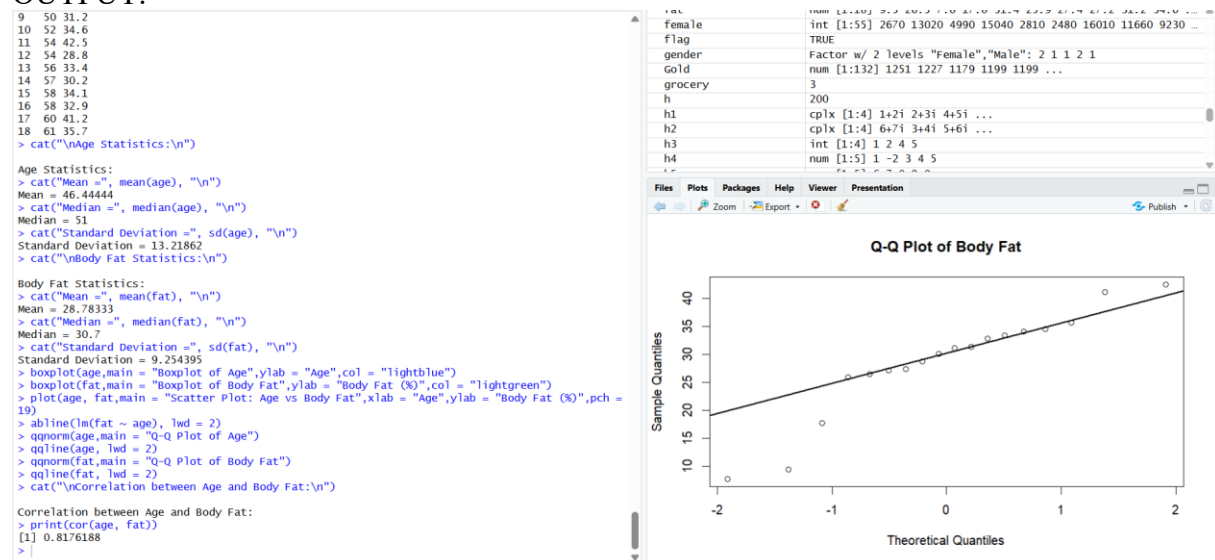
1. Enter the 18 age values.
2. Enter the corresponding 18 body-fat values.
3. Calculate the mean of age and body fat.
4. Calculate the median of age and body fat.
5. Calculate the standard deviation of age and body fat.
6. Draw a boxplot for age.
7. Draw a boxplot for body fat.
8. Draw a scatter plot of age versus body fat.
9. Draw Q-Q plots for age and body fat.
10. Calculate the correlation between age and body fat.
11. Interpret the plots and statistics.

CODE:

```
age <- c(23, 23, 27, 27, 39, 41, 47, 49, 50, 52, 54, 54, 56, 57, 58, 58, 60, 61)
fat <- c(9.5, 26.5, 7.8, 17.8, 31.4, 25.9, 27.4, 27.2, 31.2, 34.6, 42.5, 28.8, 33.4, 30.2, 34.1, 32.9, 41.2, 35.7)
data <- data.frame(age, fat)
cat("Age and Body Fat Data:\n")
print(data)
cat("\nAge Statistics:\n")
cat("Mean =", mean(age), "\n")
cat("Median =", median(age), "\n")
cat("Standard Deviation =", sd(age), "\n")
cat("\nBody Fat Statistics:\n")
cat("Mean =", mean(fat), "\n")
cat("Median =", median(fat), "\n")
cat("Standard Deviation =", sd(fat), "\n")
boxplot(age, main = "Boxplot of Age", ylab = "Age", col = "lightblue")
boxplot(fat, main = "Boxplot of Body Fat", ylab = "Body Fat (%)", col = "lightgreen")
plot(age, fat, main = "Scatter Plot: Age vs Body Fat", xlab = "Age", ylab = "Body Fat (%)", pch = 19)
abline(lm(fat ~ age), lwd = 2)
qqnorm(age, main = "Q-Q Plot of Age")
qqline(age, lwd = 2)
qqnorm(fat, main = "Q-Q Plot of Body Fat")
```

```
qqline(fat, lwd = 2)
cat("\nCorrelation between Age and Body Fat:\n")
print(cor(age, fat))
```

OUTPUT:



RESULT:

The mean age is **46.44 years** and the mean body fat is **28.78%**. The scatter plot shows a strong positive relationship between age and body fat, with correlation approximately **0.818**.